

## 9. Vegetation and Fire

The first chapters of these notes have largely covered generic subjects in passive optical remote sensing – issues of general importance and image processing techniques that are not specific to one field of application or another. In this chapter we will look at some techniques developed and widely used specifically for remote sensing of two things: vegetation, and fire. Similar chapters could be written on many other specific applications of remote sensing – vegetation and fire are used here because the use of satellite imagery to map and monitor them is well-developed and used operationally at both national and global scales.

### Vegetation

Given how much of Earth is covered by some form of green vegetation, it is not surprising that much research effort has gone into using remote sensing

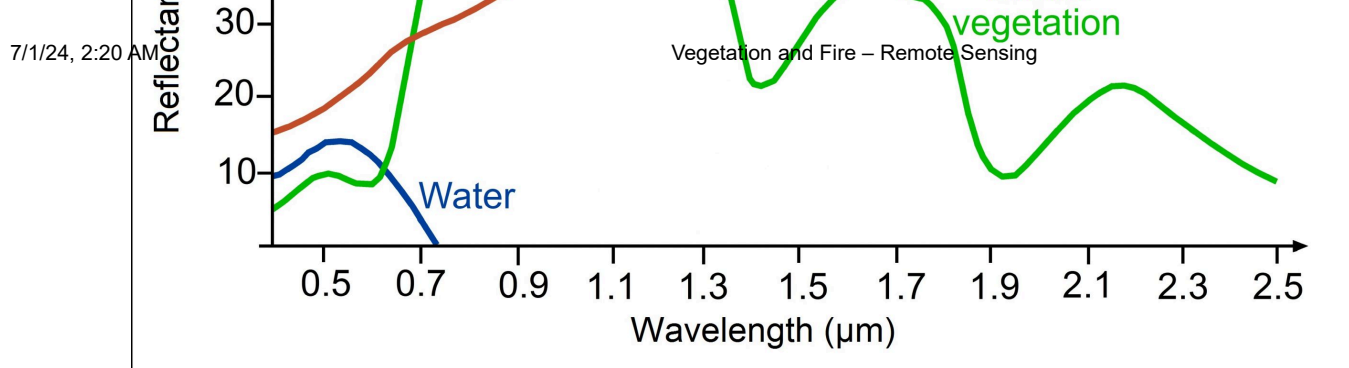


Figure 67: Generalized spectral signatures of green vegetation, soil, and water. Same as Figure 27.

Reflectance of water, soil and vegetation at different wavelengths by SEOS, Science Education through Earth Observation for High Schools (SEOS), [CC BY-NC-SA 2.0](#).

## Visible wavelength region

Healthy green vegetation absorbs most of the visible light incident on it, using it to grow through the process of photosynthesis. Photosynthesis is a relatively complex biochemical process that begins with the absorption of a photon by a pigment molecule, which sets off an electronic transition (the movement of an electron from one energy level to a higher one) that starts a chain-reaction leading to production of plant material from the basic chemical components of  $\text{CO}_2$  and water. A form of chlorophyll is the dominant pigment in almost all vegetation, and chlorophyll is very good at absorbing visible light,

parts of plants, especially liquid water found in the leaves, do absorb these photons, which rotate and stretch chemical bonds in the water or in plant cells, effectively heating up the absorbing material. In general, plants have no need for heating up, and except for wavelengths around  $1.4\ \mu\text{m}$  and  $1.9\ \mu\text{m}$  where water is a strong absorber, plants reflect much of the incoming near-infrared radiation.

The combination of low reflectance in the visible and high reflectance in the near-infrared wavelengths is the most characteristic signature of vegetation used in remote sensing, and is used for quick-and-dirty mapping of vegetation on Earth. Consider that most land areas contain a mix of vegetated and unvegetated areas. Sometimes this 'mix' is very one-sided, as in the Amazon, the Sahara desert, or the Canadian Arctic, and sometimes it is truly more mixed, as on the African savanna (Figure 68).

## Vegetation indices

Using the reflectance in the visible wavelengths alone does not tell us how much vegetation is in an area, because water also has low reflectance in this wavelength region. And using the reflectance in the near-infrared region alone also does not tell us much about vegetation, because other surface types, such as the bright sand and bright snow depicted in Figure 68 also have higher near-infrared reflectance. However, the ratio of the visible and near-infrared reflectances is a useful indicator for the amount of vegetation



Figure 68: Four areas with rather different vegetation characteristics. Top left: The African savanna contain a mix of trees and grass cover. [Trees On The African Savanna](#) by Lynn Greyling, PublicDomainPictures.net, [CC0 1.0](#). Top right: The Sahara desert is in most places void of vegetation. [Sahara Desert](#) by Azer Koçulu, Wikimedia Commons, [CC0 1.0](#). Bottom left: The Amazon rainforest is covered in dense vegetation. [Amazon Rainforest](#) by CIFOR (Neil Palmer/CIAT), Flickr, [CC BY-NC-ND 2.0](#). Bottom right: The Canadian Arctic has large areas with no vegetation. [Patterned ground in Canadian tundra](#) by Raymond M. Coveney, Wikimedia Commons, [CC BY-SA 3.0](#).

While surface reflectance values should in principle always be used to calculate this and any other vegetation index, in practice these calculations are often done instead based on TOA reflectance or even TOA radiance (or even DN values!). The Simple Ratio has one drawback that means it is rarely used – its values approach infinity when red reflectance approaches zero, which makes differences between large values difficult to interpret. A simple improvement



Figure 69: NDVI composite of Earth, showing relative differences in vegetation cover. [GlobalNdvitmo 200711 Irg](#) by Reto Stockli and Jesse Allen (NASA), Wikimedia Commons, public domain.

Vegetation indices can be used for more than producing pretty pictures of global vegetation distribution. As a result of global warming, which is especially pronounced in the Arctic region, the northern parts of Canada are undergoing rapid changes in vegetation distribution and growth patterns. Monitoring such changes over this large and very sparsely populated region is based on long-term NDVI trends in the summer months, which among other things demonstrates that region undergoing faster warming also experience more rapid growth in vegetation density. Another important use of vegetation indices is in crop yield forecasting: by tracking NDVI through an agricultural growing season, the development of the crops can be quantified and their eventual yields predicted statistically. Probably the most important use of such crop prediction models is to forecast droughts, crop failures and famines in parts of the world where people depend on local agriculture for their food supply. This is done systematically for countries that are prone to

quite clear in relative terms (higher values indicate more green vegetation), their relationship to real measurable vegetation characteristics used by people outside the field of remote sensing is not clear.

A vegetation index not intended to portray vegetation density, but rather to portray relative differences in vegetation water content, has been developed specifically to take advantage of the water absorption feature around 1.4  $\mu\text{m}$  shown in Figure 67. It is called the Normalized Difference Water Index (NDWI), and its generic equation is:

$$\text{NDWI} = (\text{NIR} - \text{SWIR}) / (\text{NIR} + \text{SWIR})$$

where SWIR denotes a measurement in the short-wave infrared region (1.4  $\mu\text{m}$ ). Dry vegetation contains less water, and therefore absorbs less of the SWIR radiation, reducing the value of NDWI.

For any vegetation index, the specific bands that are used as the 'RED', 'NIR' and 'SWIR' bands depends on the sensor in question, and not all sensors have bands appropriate for calculating one or another index.

## Mapping real vegetation attributes

Vegetation attributes that are more used by non-remote sensing people include among other things above-ground biomass, absorbed photosynthetically active radiation (APAR), and the Leaf Area Index (LAI). Above-ground biomass can be measured (by cutting and weighing it) in kg/ha, and is of fun-

A simple approach to mapping LAI is illustrated in Figure 70, in which NDVI values have been regressed against field-measured LAI. As is obvious from Figure 70, what we measure in remote sensing data is related, but not very closely, and certainly not linearly, to actual LAI measured on the ground. Nevertheless the regression lines can be used to produce a first approximation of LAI from these remote sensing data. A similar approach can be taken to map other vegetation attributes.

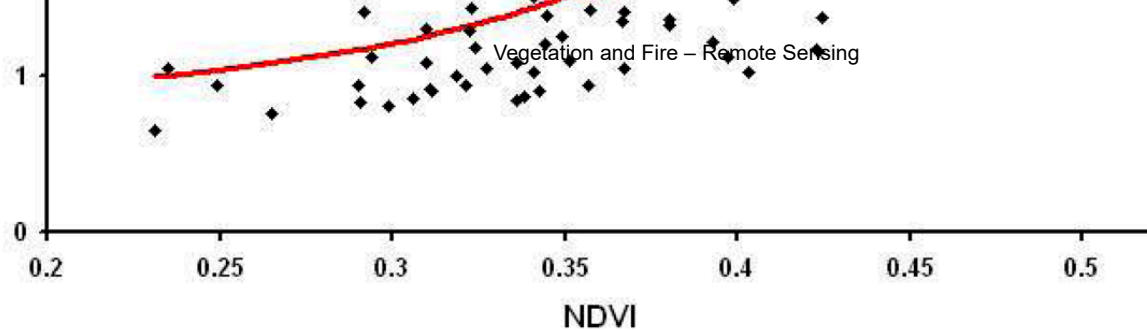


Figure 70: Empirical relationships between four vegetation indices and field-measured LAI. [Assessment of Unmanned Aerial Vehicles Imagery for Quantitative Monitoring of Wheat Crop in Small Plots](#) (Figure 8) by LeLong et al., [CC BY 3.0](#).

## Radiative transfer approach

More sophisticated approaches to mapping vegetation attributes rely on radiative transfer models. Radiative transfer models build up a simple 3D world consisting of a) the ground, b) some vegetation on the ground, c) the atmosphere, and d) the Sun. Each of these elements have specific attributes that define how they interact with radiation: The Sun emits a certain amount of radiation at each modeled wavelength, the atmosphere absorbs and scatters this radiation to some degree, and the vegetation and the ground absorb and reflect the radiation. Once specified, the model can then be 'run' to determine

7/1/24, 2:20 AM Vegetation and Fire – Remote Sensing

the radiative transfer model to determine what their specific combination would look like at the top of the atmosphere, allows you to build up a look-up table with a) environmental conditions including the parameter of interest (e.g. LAI) and b) the resulting TOA (spectral) radiance. When mapping LAI with a satellite image, the observed TOA radiance of each pixel can then be matched to the entry in the table it matches most closely to determine the probably value of LAI in that pixel.

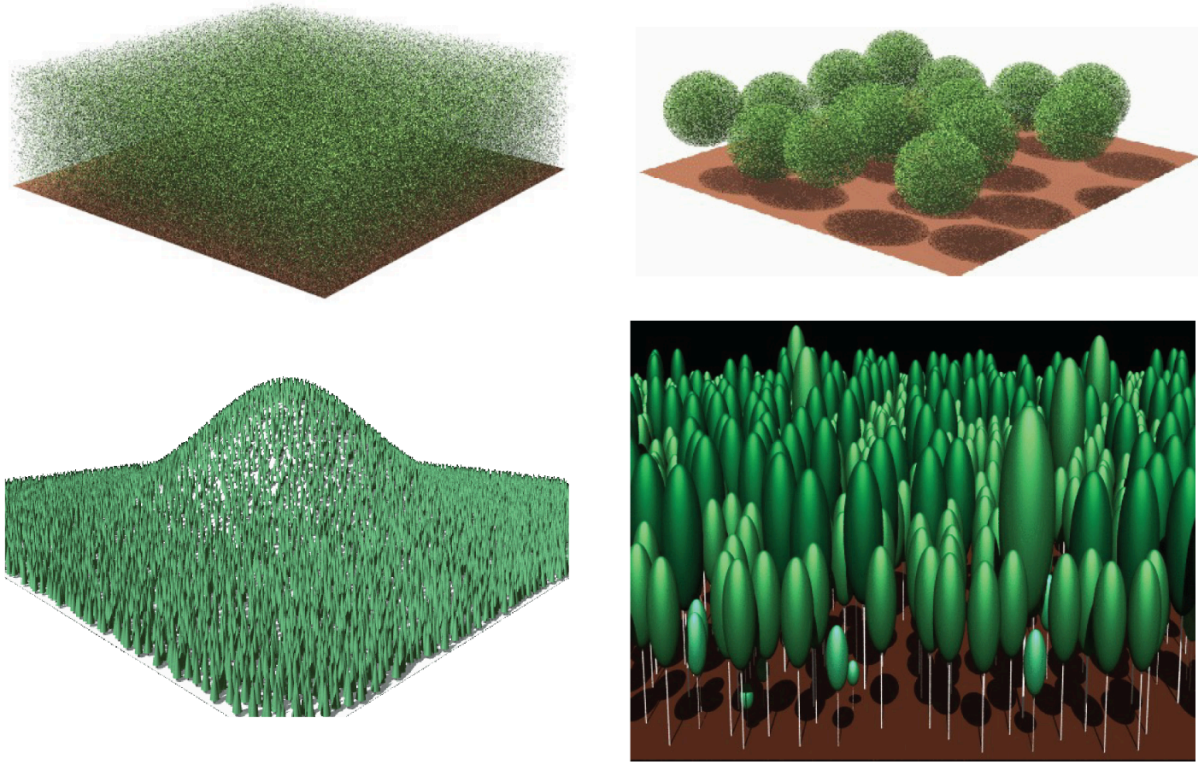


Figure 71: Example of a radiative transfer model of vegetation. Modified from [Accelerated RAPID Model Using Heterogeneous Porous Objects](#) (Figure 7) by Huaguo Huang, [CC BY 4.0](#).



Figure 72: Example of active fire detection from space during the Australian wildfires of 2019. [Satellite image of bushfire smoke over Eastern Australia](#) by the European Space Agency, Flickr, [CC BY-SA 2.0](#).

Once a fire is over, it may be important to assess how much area was burned, and how intensely it was burned. This has important implications for the recovery of the vegetation and associated biota in the area, and it can also help us answer important questions related to climate change, like how the fre-

ages, one from before and one from after the fire, which leads to the calculation of  $\delta\text{NBR}$ :

$$\delta\text{NBR} = \text{NBR}_{\text{before}} - \text{NBR}_{\text{after}}$$

To minimize the effects of other environmental factors on this calculation, it is ideal to use ‘anniversary dates’ to calculate  $\delta\text{NBR}$ , although this is not necessary if two images are available from a small period covering just before and just after the fire (Figure 73).

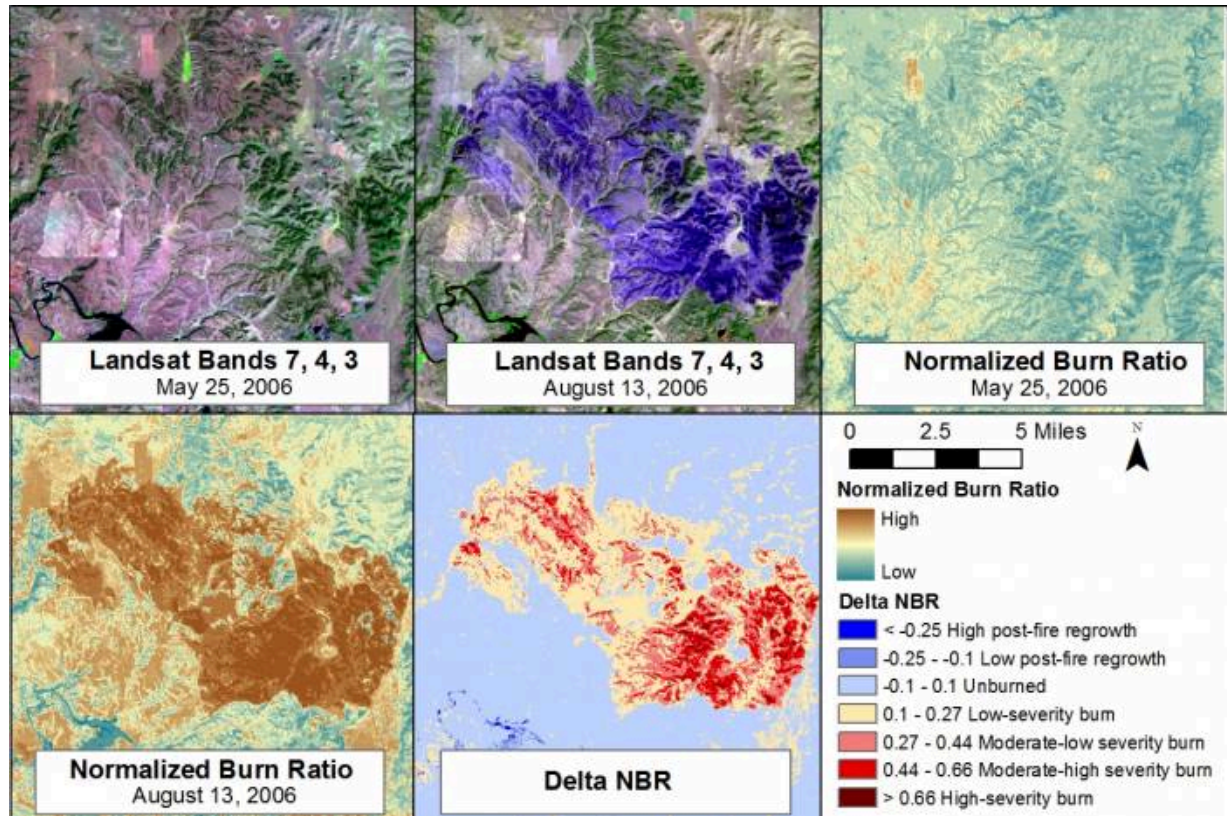




Figure 74: An example of both vegetation and fire, both of which can be mapped well with remote sensing. [Forest Fire](#), by Ylvers, Pixabay, [Pixabay License](#).

## LICENSE



## SHARE THIS BOOK



